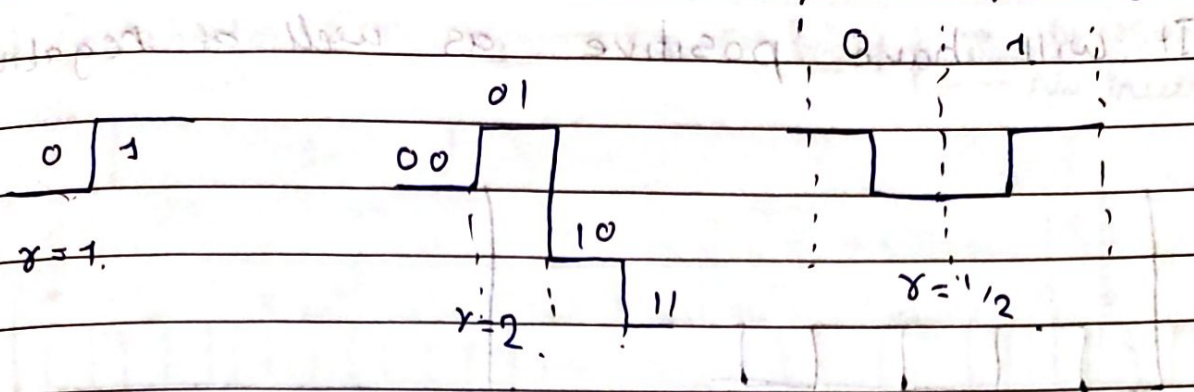


$r = 2/1 \Rightarrow 2$ data elements provided per Signal element



$r = \frac{3}{4}$ $2^3 = 8$ $2^4 = 16$

we are using 8 patterns out of 16

$S = \frac{N}{r}$ — Signal rate

* Line Coding Algorithm:

Conversion of 0s and 1s into digital Signal is called line coding

* Baseline Wandering effect: (Should be less)

* Self Synchronization

$\rightarrow$ Clock Speed difference in Receiver & Transmitter.

* DC Components: (There should be minimum DC Components)

* Complexity - Algorithm must be algorithm

* Built in error detection

* Immunity to noise interference

Algorithms

* Unipolar NRZ (Non return to Zero)

$S_{avg} = C \times N \times \frac{1}{r}$

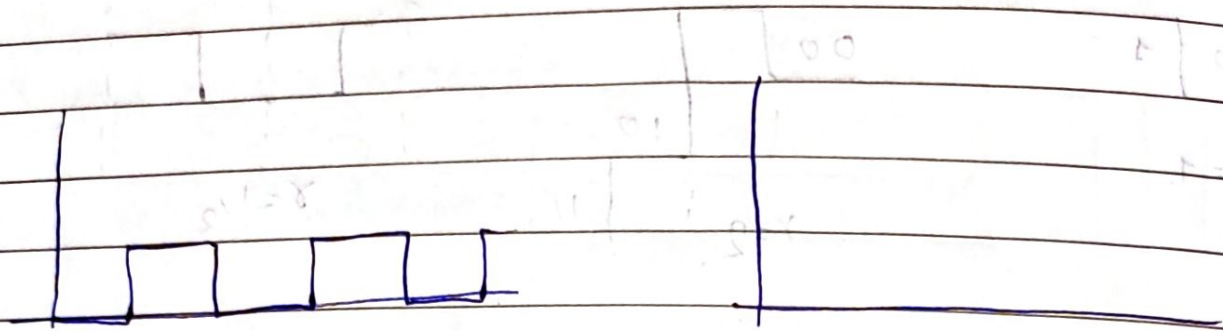
Cost factor

$N_{max} = \frac{1}{C} \times S \times r$

$= 2 \times B \times \log_2 L$
 $\Rightarrow C = \frac{1}{2}$

* Polar NRZ

It will have positive as well as negative



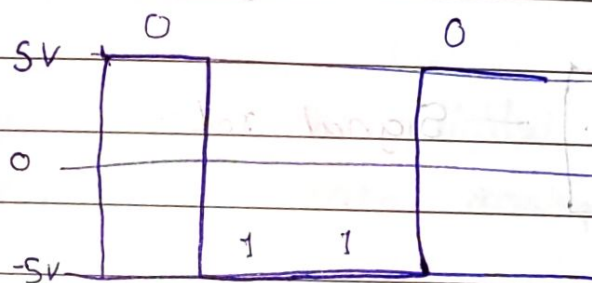
Unipolar

One of the algorithm in Polar NRZ is

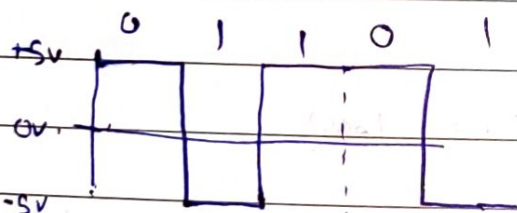
→ Polar NRZ Level.

0 = +5V

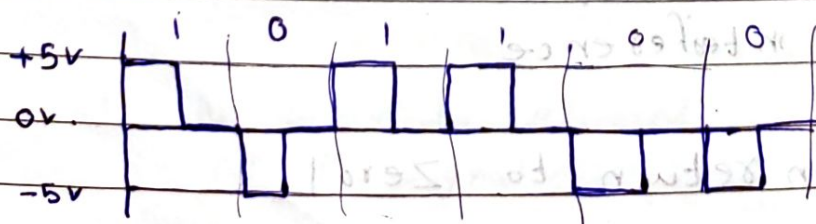
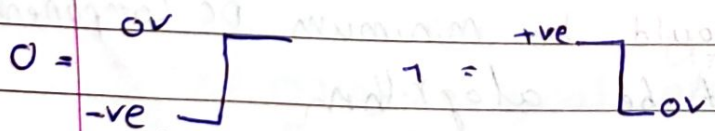
1 = -5V



→ Polar NRZ I



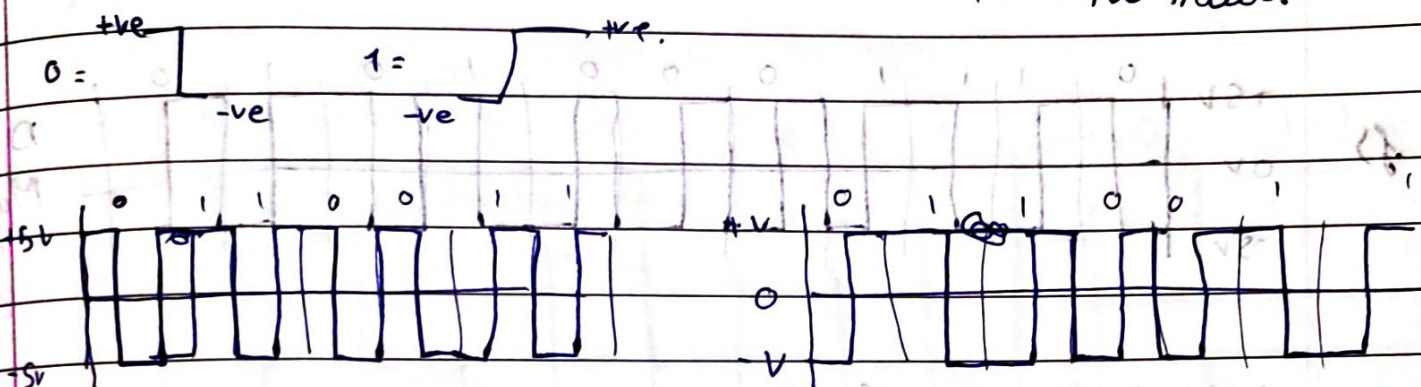
* Polar RZ (Return to Zero)



What is its
bandwidth
bitrate etc

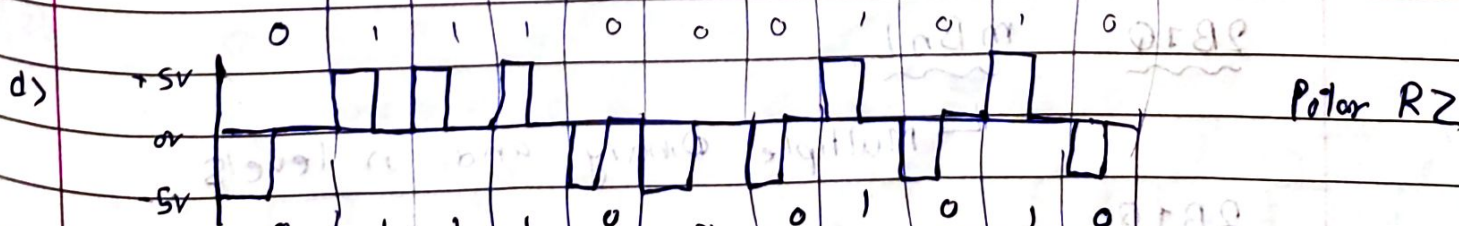
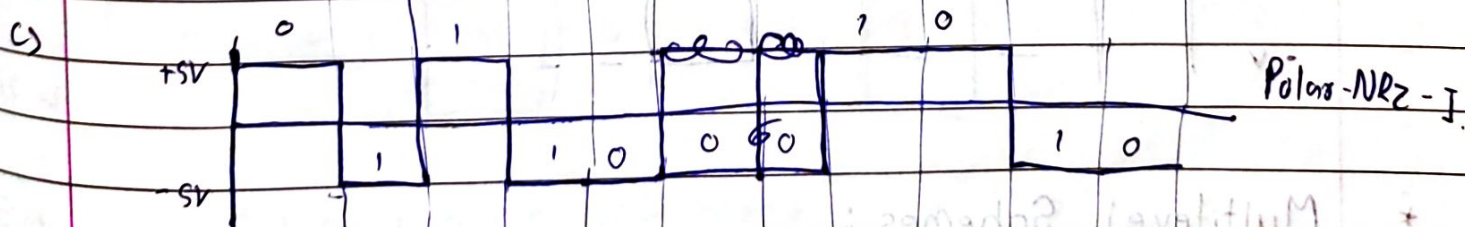
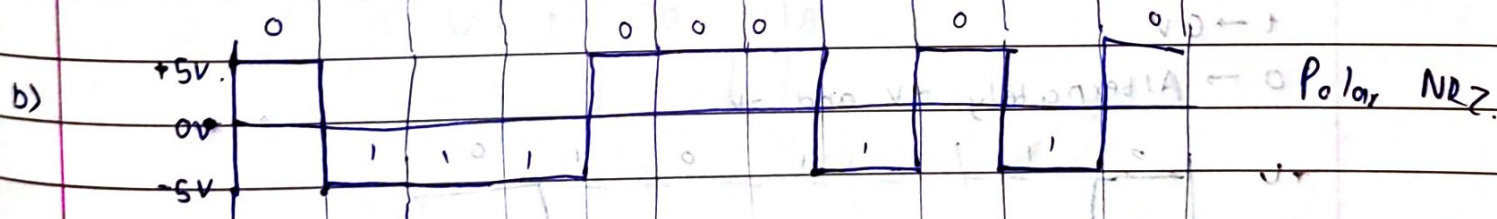
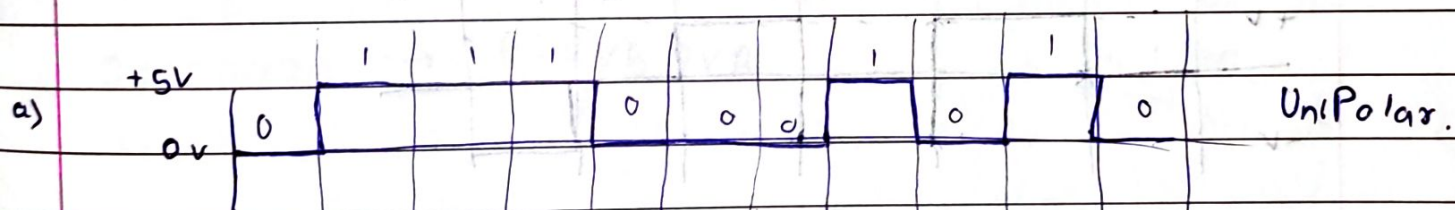
Manchester Encoding

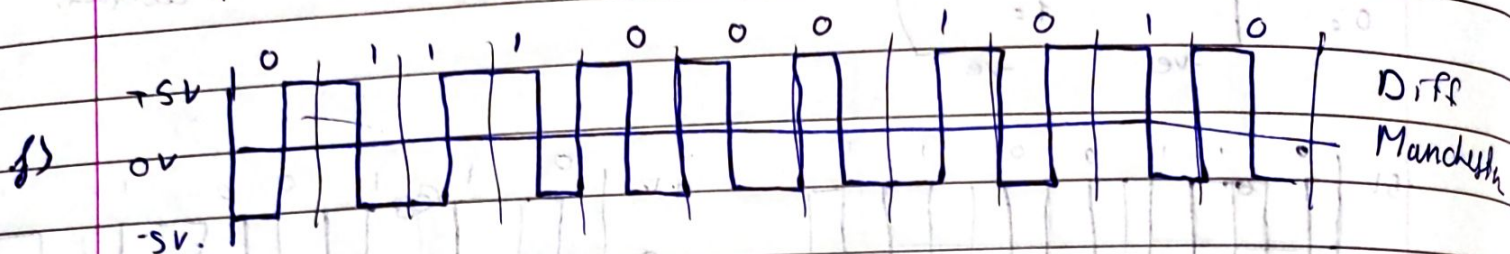
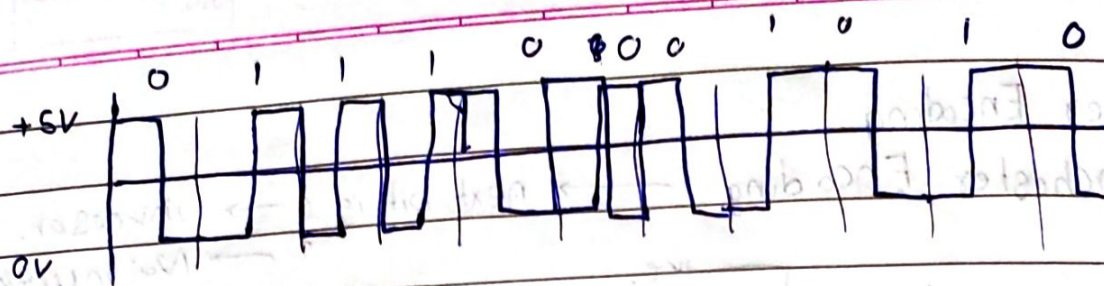
Diff Manchester Encoding $\rightarrow$ next bit is 0 $\rightarrow$ Inversion.
 $1 \rightarrow$ No Inversion.



Sketch the following signals for the given binary input $\rightarrow 01110001010$

- a) Unipolar - NRZ b) Polar NRZ-L c) Polar -NRZ-I
 d) Polar -RZ f) Diff Manchester e) Manchester



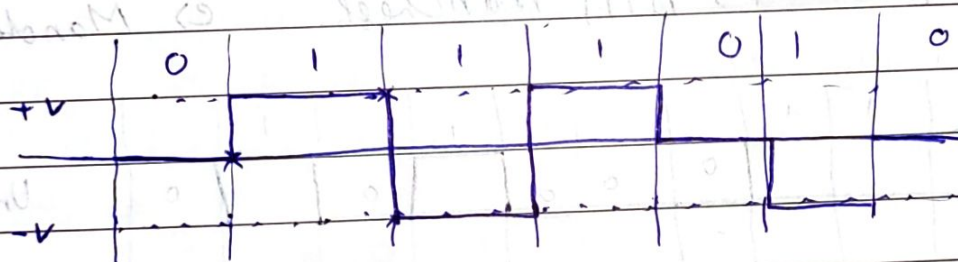


* AMI & Pseudoternary

AMI stands for alternate mark inversion.

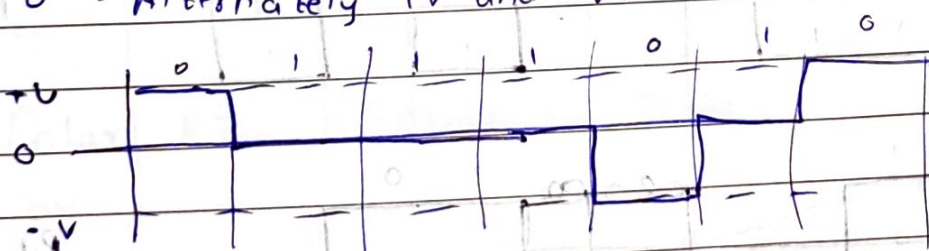
0 $\rightarrow$ 0V.

1 $\rightarrow$ Alternately +V or -V.



1 $\rightarrow$ 0V

0 $\rightarrow$ Alternately +V and -V.



* Multilevel Schemes:

2B1Q

mBnL

$\rightarrow$ Multiple binary and n levels

2B1Q:

$\rightarrow$

2

binary

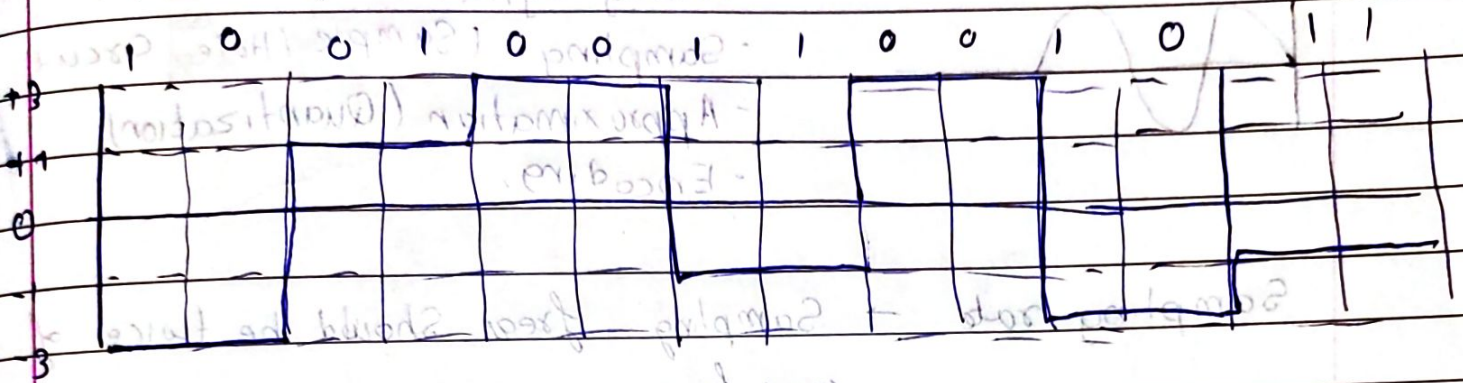
1 Quaternary

00 - +3

01 - +1

10 - -3

11 - -1



* Scrambling:

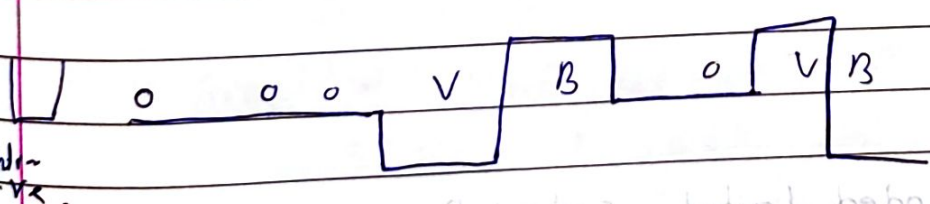
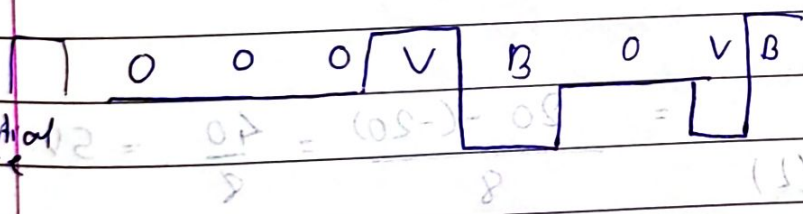
B8ZS.

Bipolar with 8 Zero Substitution

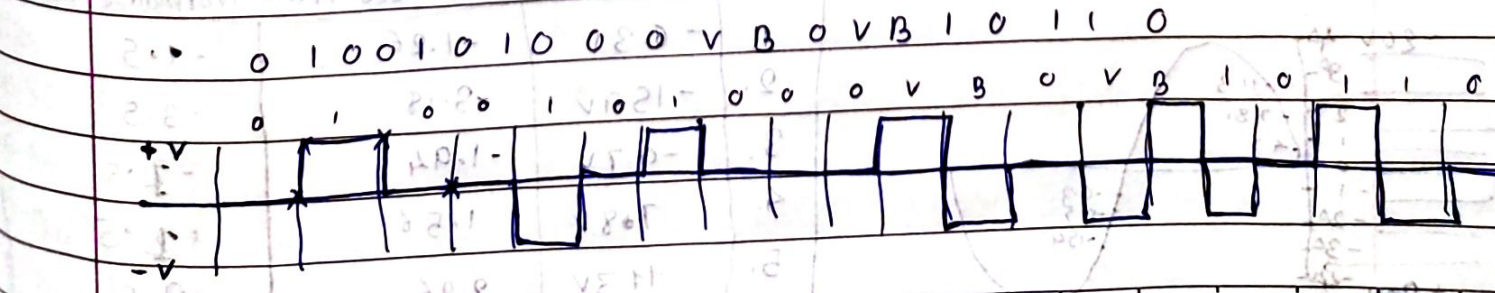
It use alternate mark inversion (Modified)

00000000 → 000VB0VB

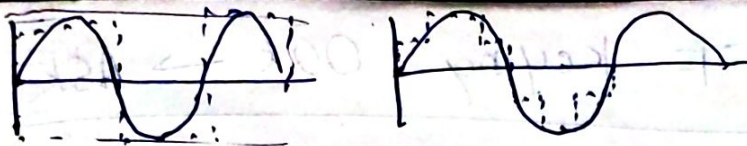
Violation Bipolar.



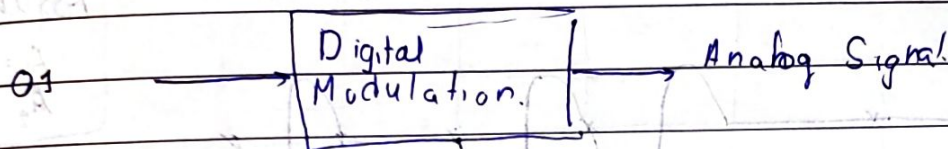
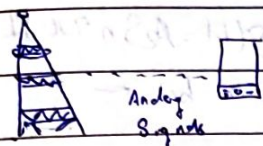
Find the encoded digital code for 011001010000000010110



PCM → Pulse Code Modulation.



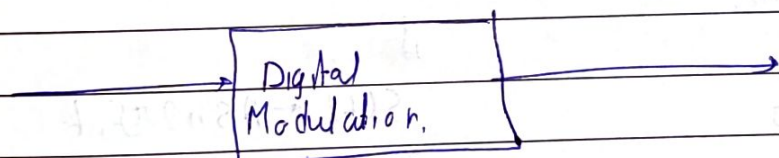
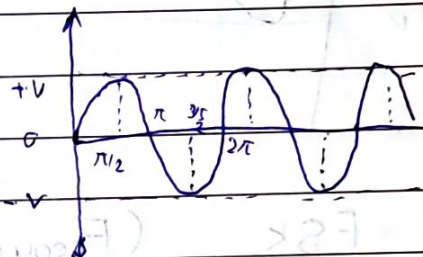
* Digital to Analog Conversion :



$$S(t) = A \sin(\omega t + \phi)$$

$$= A \sin(2\pi f t + \phi)$$

Amplitude Frequency Phase



ASK - Amplitude Shift Keying
 FSK - Frequency Shift Keying
 PSK - Phase Shift Keying

1) Amplitude Shift Keying

$$S(t) = \begin{cases} A_1 \sin 2\pi f t & \text{for } 1 \\ 0 & \text{for } 0 \end{cases}$$

Quantization Error	Quantization Code	Encoded Word
0.24	2	010
-0.32	0	000
-0.56	2	010
0.06	5	101
0.24	6	110

ON-OFF keying

if $1/p \leftarrow 0$

$$s(t) = A \sin 2\pi f t$$

else if

$$s(t) = A_2 \sin 2\pi f t$$

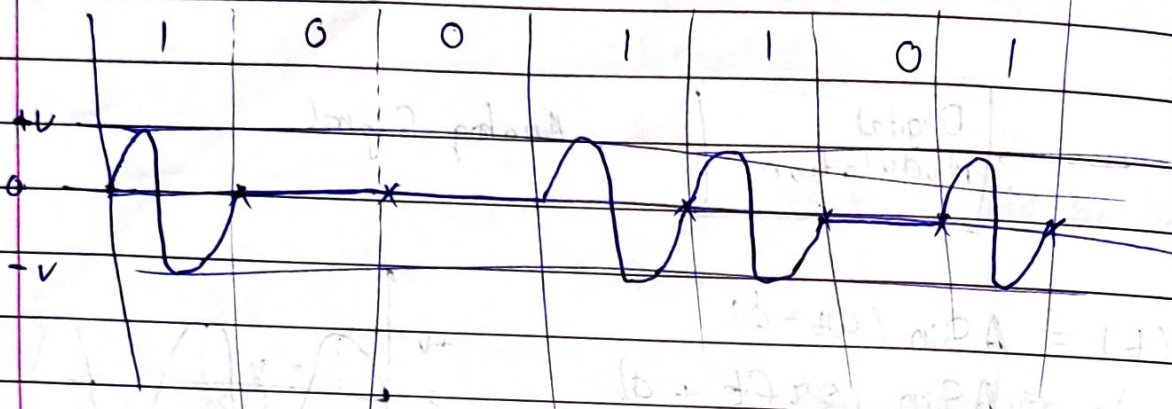
$$s(t) = A_1 \sin 2\pi f t$$

then bit $\leftarrow 0$

else if

$$s(t) = A_2 \sin 2\pi f t$$

then bit $\leftarrow 1$



* FSK (Frequency Shift Keying)

$\phi = 0$ $A = 2V$ f_1, f_2

Scan 1p.

if $1/p \leftarrow 0$

$$s(t) = A \sin 2\pi f_1 t$$

else $1/p \leftarrow 1$

$$s(t) = A \sin 2\pi f_2 t$$

if

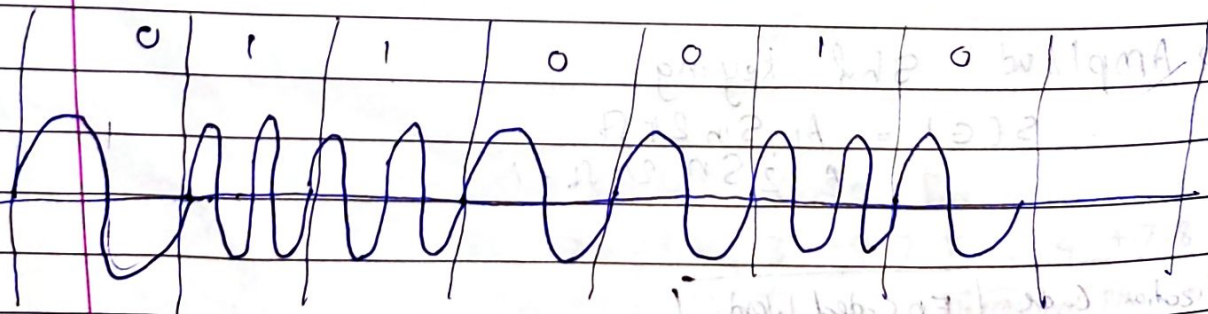
$$s(t) = A \sin 2\pi f_1 t$$

bit $\leftarrow 0$

else if

$$s(t) = A \sin 2\pi f_2 t$$

bit $\leftarrow 1$



* BPSK (Binary Phase Shift Keying)

$$s(t) = A \sin (2\pi f t + \phi_1), 0$$

Scan input.

if $i/p \leftarrow 0$

$$S(t) = A \sin 2\pi(f t + \phi_1)$$

else

$$S(t) = A \sin 2\pi f t$$

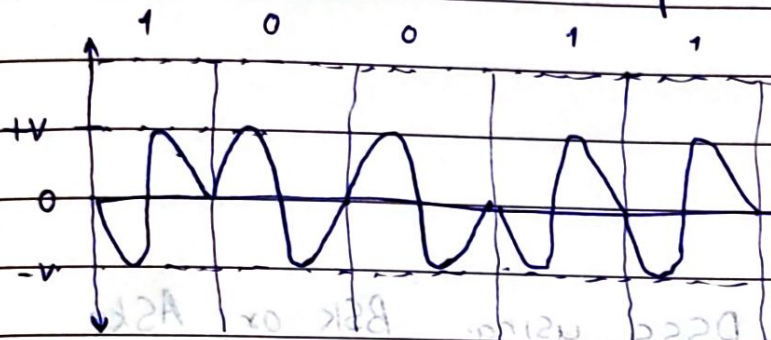
end

$$S(t) \leftarrow A \sin 2\pi f t + \phi_1$$

bit $\leftarrow 0$

else

bit $\leftarrow 1$



$$\text{BaudRate} = \text{Bitrate} \quad (\text{ASK, FSK, BPSK})$$

QPSK (Quadrature Phase shift keying)

$$A \sin (2\pi f t + \phi_1) \quad 00$$

$$A \sin (2\pi f t + \phi_2) \quad 01$$

$$A \sin (2\pi f t + \phi_3) \quad 10$$

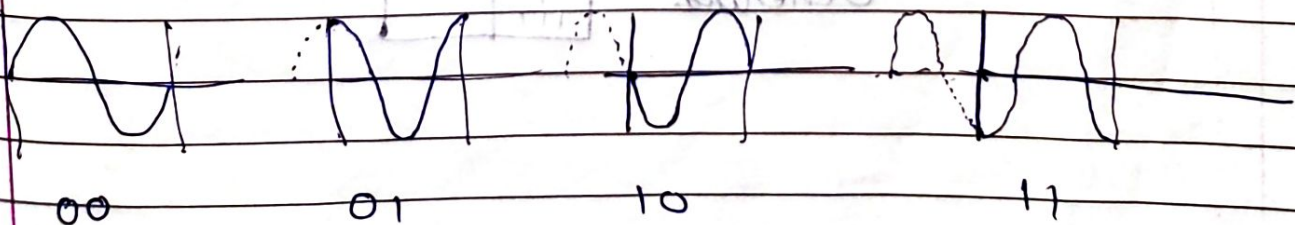
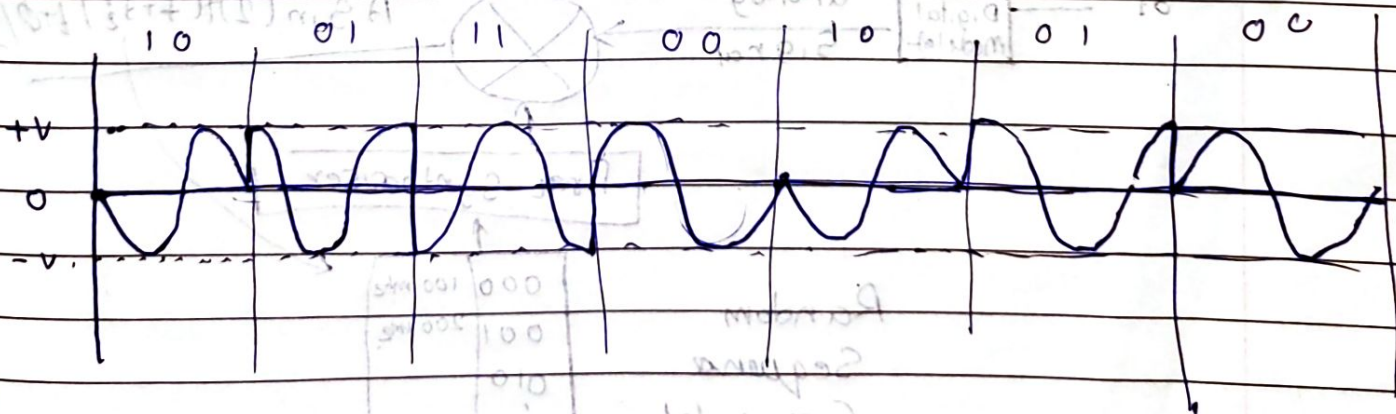
$$A \sin (2\pi f t + \phi_4) \quad 11$$

$$\phi_1 = 0^\circ = 0$$

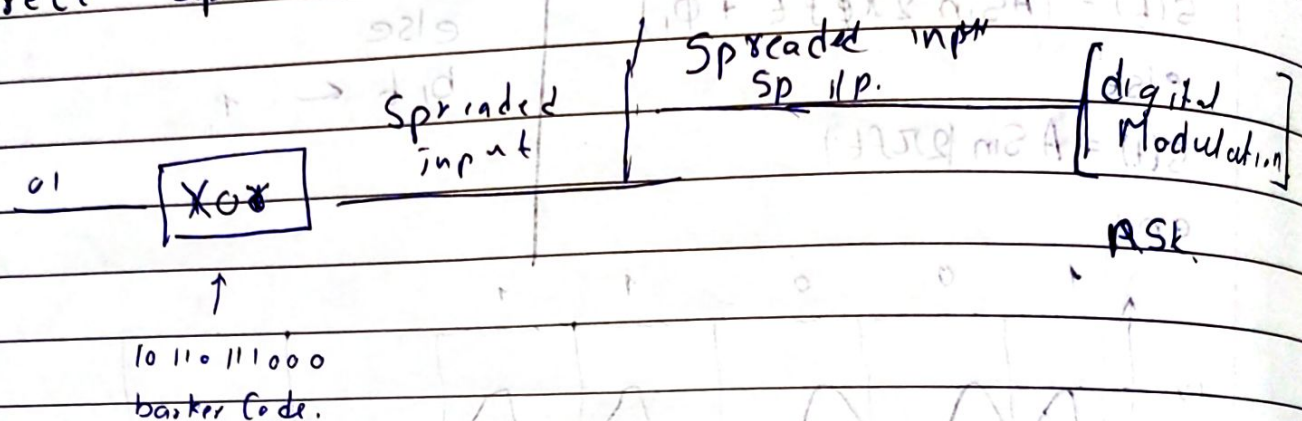
$$\phi_2 = 90^\circ = \pi/2$$

$$\phi_3 = 180^\circ = \pi$$

$$\phi_4 = 270^\circ = 3\pi/2$$



Speed Spectrum Direct Sequence Spread Spectrum (DSSS)



0110 Sketch DSSS using BSK or ASK

* Frequency hopping Spread Spectrum

* Dwell time - The amount of time a channel stays on a frequency.

* Hopping time.

$$s(t) = A \sin(2\pi f_c t + \phi)$$

